

Probability and Statistics

Lecture 1 - Probability and Algebra of Probability

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Introduction

Two fundamental ways of defining probability:

- ▶ Classical (theoretical) approach
what should happen theoretically
- ▶ Statistical (empirical) approach
what actually happens in practice

Classical approach

- ▶ In this way, probability is based on theory and logical reasoning. It assumes that all outcomes are equally likely.

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

- ▶ Classical approach:
 - ▶ Is based on reasoning, not experimentation
 - ▶ Works only when outcomes are equally likely
 - ▶ Requires when outcomes are known in advance
 - ▶ Mostly used in games of chances (cards, dice, coins)

[Note: Equally likely outcomes means that each possible outcome has the same chance of occurring]

Classical approach

► Examples:

► When tossing a fair coin:

- Total outcomes = 2 (Head, Tail)
- Probability of Head = $1/2$

► When rolling a fair die:

- Total outcomes = 6 (1, 2, 3, 4, 5, 6)
- Probability of getting multiples of 3 = $2/6$

Limitation: This way of getting probability is not suitable when

- outcomes are not equally likely
- the total possible outcomes are unknown

Statistical approach

- ▶ In this way, probability is defined based on actual experiments or past data.

$$P(E) = \frac{\text{Number of times event occurs}}{\text{Total number of trials}}$$

- ▶ Statistical approach:
 - ▶ Is based on observation and experimentation
 - ▶ Also called relative frequency approach
 - ▶ More practical for real-life situations
 - ▶ Does not require equally likely outcomes
 - ▶ Mostly used in weather, business, medicine

Statistical approach

► Examples:

► If it rained 13 days out of 50 observed days:

$$P(Rain) = \frac{13}{50} = 0.26$$

Limitation:

- probability may change with more data
- it requires a large number of trials for accuracy

Probability - Terminologies

Term	Definition	Examples
Random Experiment	An experiment: 1. that has more than one outcome 2. whose result cannot be predicted exactly before it happens, even though it is repeated under similar conditions.	1. Tossing a coin 2. Rolling a die 3. Drawing a card from a deck. 4. Selecting a student at random from the class.
Sample Space	A set containing all the possible outcomes of a random experiment is called Sample Space for that experiment and is denoted by S. Each element in Sample Space (S) is called Sample Point.	1. tossing a coin $S=\{H, T\}$ 2. tossing two coins $S=\{HH, HT, TH, TT\}$ 3. Roll a die $S=\{1, 2, 3, 4, 5, 6\}$ 4. Roll two dice $S=\{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

Probability - Terminologies

Term	Definition	Examples
Event	A subset of sample space is called an Event.	Tossing two coins: $S = \{HH, HT, TH, TT\}$ E_1 = a HEAD appears = $\{HT, TH\}$ E_2 = at least one head appear = $\{HT, TH, HH\}$ E_3 = at most two tail appear = $\{HH, HT, TH, TT\}$
Occurrence of an Event	Let E is an event then we say that E occurs if the outcome $x \in S$ in experiment such that $x \in E$. An event occur if it happens	E = Even number on a die = $\{2, 4, 6\}$ If you roll a die and you get 2, we say the event E has occurred. A = a number less than 4 = $\{1, 2, 3\}$ If you roll a die and obtain 1, 2 or 3, you can say that event A has occurred. And if the outcome is 2, you can say that both events E and A occurred.

Types of Events

Name	Definition	Examples
Simple Event	Any event is said to be simple if it has only one sample point of a Sample space.	$S = \{1, 2, 3, 4, 5, 6\}$ $E_1 = \{1\}$ $E_2 = \{2\}$ $E_3 = \{3\}$
Compound Event	Any event is said to be Compound if it contains more than one sample points of a sample space.	If we toss a coin twice: $S = \{HH, HT, TH, TT\}$ $A = \text{at least one HEAD} = \{HT, TH, HH\}$ $B = \text{at most one TAIL} = \{HT, TH, HH\}$
Sure (Certain) Event	Any event is said to be sure if $A = S$ i.e. no missing outcome in A.	$A = \text{a number from 1 to 6 on a die}$

Types of Events

Name	Definition	Examples
Impossible Event	Any event is said to be impossible if $A = \phi$, i.e. no outcome in A.	A = a number greater than 6 on a die If you roll a die, you never obtain a number greater than 6. So, A is an impossible event.
Complementary Events	Two events where one occurs if and only if the other does not occur.	If rolling a die: A =getting an even number A' =getting an odd number
Mutually Exclusive Events	Events that cannot occur at the same time	$A = \{1\}$ $B = \{2\}$ You cannot get both 1 and 2 in one roll.

Complementary events together cover all possible outcomes... $P(A) + P(B) = 1$
Mutually exclusive events do not cover all outcomes... $P(A) + P(B) \leq 1$

Types of Events

Name	Definition	Examples
Non-Mutually Exclusive Events	Events that can occur together	A = even number {2, 4, 6} B = number greater than 3 {4, 5, 6} You can get 4 or 6 which are common outcomes in both events
Independent Events	Two events where the occurrence of one does not affect the other.	Tossing a coin and rolling a die
Dependent Events	Two events where the occurrence of one affects the probability of the other.	Drawing two cards without replacement
Exhaustive Events	A set of events is exhaustive if at least one of them must happen —in other words, they cover all possible outcomes of an experiment.	Roll a six-sided die: S: {1, 2, 3, 4, 5, 6} Events: A1 = {1, 2} A2 = {3, 4} A3 = {5, 6}

Algebra of Events

► Complementary Events

If A is an event, then A' is complementary of A

$$A' = \{x \mid x \in S, x \notin A\}$$

Example: A = getting an even number = $\{2, 4, 6\}$

A' = not A = getting odd no = $\{1, 3, 5\}$

$$A \cup A' = S$$

Algebra of Events

► Event A or B

If A is an event and B is another event, then

Event A or B means

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Example: A = an even number = {2, 4, 6}

B = no less than 5 = {1, 2, 3, 4}

$$A \cup B = \{1, 2, 3, 4, 6\}$$

If you get any number from $A \cup B$, it means either event A occurred (like 6), or event B occurred (like 1,3), or both occurred (like 2,4)

Algebra of Events

► Event A and B

If A is an event and B is another event, then
Event A and B means

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example: A = an even number = {2, 4, 6}

B = no less than 5 = {1, 2, 3, 4}

$$A \cap B = \{2, 4\}$$

If you get any number from $A \cap B$, it means both events occurred together

Algebra of Events

► Event A but not B

If A is an event and B is another event, then

Event A but not B means

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

Example: A = an even number = {2, 4, 6}

B = no less than 5 = {1, 2, 3, 4}

$$A - B = \{6\}$$

If you get any number from $A - B$, it means event A occurs but event B does not.

Algebra of Events

► Mutually Exclusive Event

If A is an event and B is another event, then they are called mutually exclusive if

$$A \cap B = \phi$$

For events A, B, C: $A \cap B = \phi$, $B \cap C = \phi$, $A \cap C = \phi$

Example: A = an even number = {2, 4, 6}

B = number less than 5 = {1, 3, 5}

$$A \cap B = \phi$$

it means both events A and B cannot occur at the same time.

Algebra of Events

► Exhaustive Event

If union of 2 or more events is equal to S , they are called exhaustive events

$$A \cup B \cup C = S$$

Example: $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5, 6\}$

$$A \cup B \cup C = S$$

it means at least one of the events must occur.

Exercise-1

A die is rolled. Let A be the event “Die shows 4” and B be the event “Die shows even number”. Are A and B mutually exclusive?

$$A = \{4\}$$

$$B = \{2, 4, 6\}$$

$$A \cap B = \{4\} \text{ that is not } \phi$$

So, A and B are not mutually exclusive events.

Exercise-2

A die is rolled. Let A be the event “Die shows 4” and B be the event “Die shows even number”. Are A and B exhaustive events?

$$A = \{4\}$$

$$B = \{2, 4, 6\}$$

$$A \cup B = \{2, 4, 6\} \text{ that is not } S$$

So, A and B are not exhaustive events.

Exercise-3

A die is rolled. The sample space $S = \{1, 2, 3, 4, 5, 6\}$

Describe the following events:

$A =$ A number less than 7 $= \{1, 2, 3, 4, 5, 6\}$

$B =$ A number greater than 7 $= \phi$

$C =$ A multiple of 3 $= \{3, 6\}$

$D =$ A number less than 4 $= \{1, 2, 3\}$

$E =$ An even number greater than 4 $= \{6\}$

$F =$ A number not less than 3 $= \{3, 4, 5, 6\}$, $F' = \{1, 2\}$

Also find $A \cup B$, $A \cap B$, $B \cup C$, $E \cap F$, $D \cap E$, $A - C$, $D - E$, $E \cap F'$

Exercise-3

A die is rolled. The sample space $S = \{1, 2, 3, 4, 5, 6\}$

Describe the following events:

$A = \text{A number less than 7} = \{1, 2, 3, 4, 5, 6\}$ (sure event)

$B = \text{A number greater than 7} = \phi$ (impossible event)

$C = \text{A multiple of 3} = \{3, 6\}$

$D = \text{A number less than 4} = \{1, 2, 3\}$

$E = \text{An even number greater than 4} = \{6\}$

$F = \text{A number not less than 3} = \{3, 4, 5, 6\}$, $F' = \{1, 2\}$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$B \cup C = \{3, 6\}$$

$$D \cap E = \phi$$

$$D - E = \{1, 2, 3\}$$

$$A \cap B = \phi$$

$$E \cap F = \{6\}$$

$$A - C = \{1, 2, 4, 5\}$$

$$E \cap F' = \phi$$

Exercise-3

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$A \cap B = \phi$$

Mutually exclusive

$$B \cup C = \{3, 6\}$$

$$E \cap F = \{6\}$$

$$D \cap E = \phi$$

Mutually exclusive

$$A - C = \{1, 2, 4, 5\}$$

$$D - E = \{1, 2, 3\}$$

$$E \cap F' = \phi$$

Mutually exclusive

Probability

- Every random experiment has a sample space:

$$S = \{x_1, x_2, x_3, \dots, x_n\}$$

- $A \subseteq S$ means A is an event, $A = \{x_1, x_2, \dots, x_i\} \ i \leq n$

$$P(A) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of possible outcomes}}$$

$$P(A) = \frac{n(A)}{n(S)},$$

where each sample point (x_i) is equally likely

Probability - Example

- Tossing two coins has a sample space:

$$S = \{HH, HT, TH, TT\}, \quad n(S) = 4$$

- A = at least one head comes = $\{HH, HT, TH\}$, $n(A) = 3$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{4}$$

Probability - important points

► Let A is any event and $A \subseteq S$

1. $0 \leq P(A) \leq 1$

2. $x_1, x_2, x_3, \dots, x_n \in S$

3. If $x_1, x_2, x_3, \dots, x_m \in A$

$P(\phi) = 0$ and $P(S) = 1$

then $P(x_1) + P(x_2) + P(x_3) + \dots + P(x_n) = 1$

then $P(x_1) + P(x_2) + P(x_3) + \dots + P(x_m) = P(A)$
where $m \leq n$

Probability - Example

- Tossing two coins has a sample space:

$$S = \{HH, HT, TH, TT\}$$

- A = at least one head comes = $\{HH, HT, TH\}$

$$P(HH) = \frac{n(A)}{n(S)} = \frac{1}{4}, \quad P(HT) = \frac{1}{4}, \quad P(TH) = \frac{1}{4}$$

$$P(HH, HT, TH) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \quad \text{OR} \quad P(A) = \frac{n(A)}{n(S)} = \frac{3}{4}$$

Probability of Event 'A or B'

$A \cup B$ = occurrence of A or B or Both

$P(A \cup B)$ = Probability of occurrence of A or B or Both

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: Tossing two coins has a sample space:

$$S = \{HH, HT, TH, TT\}$$

Event A: at least one Head occurs = $\{HT, TH, HH\}$

Event B: at least one Tail occurs = $\{HT, TH, TT\}$

But $A \cap B = \{HT, TH\}$, whereas $A \cup B = \{HT, TH, HH, TT\}$

$$P(A \cup B) = \frac{n(A \cup B)}{n(S)} = \frac{4}{4} = 1$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)} = \frac{3}{4} + \frac{3}{4} - \frac{2}{4} = \frac{3+3-2}{4} = 1 \end{aligned}$$

Probability of Event 'A or B'

If $A \cap B = \phi$ (i.e. A and B are mutually exclusive),
then

$$P(A \cap B) = 0$$

therefore,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - 0$$

$$P(A \cup B) = P(A) + P(B)$$

Probability of Event 'not A'

Let S = Sample space and A is any event

$$P(S) = 1, P(A) = \frac{n(A)}{n(S)}$$

A' = Complement of A = 'Event not A '

$$A' = S - A$$

$$P(A') = P(S) - P(A)$$

$$P(A') = 1 - P(A)$$

$$P(\text{not } A) = P(S) - P(A)$$

Probability of Event 'not A'

$$P(\text{not } A) = P(S) - P(A)$$

Example: $S = \{1, 2, 3, 4, 5, 6\}$

Event A: outcome is even number = $\{2, 4, 6\}$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

$$P(A') = 1 - P(A) = 1 - \frac{1}{2} = \frac{2-1}{2} = \frac{1}{2}$$

$$A' = S - A = \{1, 3, 5\}$$

$$P(A') = \frac{n(A')}{n(S)} = \frac{3}{6} = \frac{1}{2}, \text{ hence proved.}$$

Probability of Event not 'A or B'

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P[(A \cup B)'] = ???$$

Apply De Morgan's Law

$$P[(A \cup B)'] = P(A' \cap B')$$

Recall: Permutation and Combination

Permutation is a way to arrange given objects in different ways, **where order matters**.

$$P(n, r) = \frac{n!}{(n-r)!}$$

Combination is a way of choosing items where **order does not matter**.

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Example-1:

Which of the following cannot be a valid assignment of probabilities for outcome of a sample space $S=\{w_1, w_2, w_3, w_4, w_5, w_6, w_7\}$?

To identify this, make sure

(1) $0 \leq P(w_i) \leq 1$, (2) $\sum P(w_i) = 1$

- a. $P(w_1)=\frac{1}{7}$, $P(w_2)=\frac{1}{7}$, $P(w_3)=\frac{1}{7}$, $P(w_4)=\frac{1}{7}$,
 $P(w_5)=\frac{1}{7}$, $P(w_6)=\frac{1}{7}$, $P(w_7)=\frac{1}{7}$ (valid)
- b. $P(w_1)=P(w_2)=P(w_3)=0.14$, $P(w_4)=P(w_5)=-0.14$,
 $P(w_6)=P(w_7)=\frac{1}{7}$ (invalid)

Example-1:

- c. $P(w_1)=0.1, P(w_2)=0.2, P(w_3)=0.3, P(w_4)=0.4,$
 $P(w_5)=0.5, P(w_6)=0.6, P(w_7)=0.7$ (invalid)
- d. $P(w_1)=P(w_2)= 0.16, P(w_3) = 0.15, P(w_4)=P(w_5) =$
 $0.12, P(w_6) =0.14, P(w_7) =0.15$ (valid)
- e. $P(w_1)= 0.25, P(w_2)=P(w_3)=P(w_4) = 0, P(w_5) = 0.5,$
 $P(w_6)=P(w_7) = 0.125$ (valid)
- f. $P(w_1)=\frac{1}{7}, P(w_2)=P(w_3)=P(w_4) = 0, P(w_5) = \frac{9}{7},$
 $P(w_6)=P(w_7) = \frac{3}{7}$ (invalid)

Example-2:

A die is thrown, find the probability of following events.

- a. A prime number will appear
- b. A number greater than or equal to 3 will appear
- c. A number less than or equal to 1 will appear
- d. A number more than 6 will appear
- e. A number less than 6 will appear

Example-2:

Let $S = \{1, 2, 3, 4, 5, 6\}$, $n(S)=6$

a. A prime number will appear

$$E = \{2, 3, 5\}, n(E)=3$$

$$P(E) = \frac{3}{6} = \frac{1}{2}$$

b. A number greater than or equal to 3 will appear

$$E = \{3, 4, 5, 6\}, n(E)=4$$

$$P(E) = \frac{4}{6} = \frac{2}{3}$$

Example-2:

- c. A number less than or equal to 1 will appear

$$E = \{1\}, n(E)=1$$

$$P(E) = \frac{1}{6}$$

- d. A number more than 6 will appear

$$E = \{ \}, n(E) = 0$$

$$P(E) = \frac{0}{6} = 0$$

- e. A number less than 6 will appear

$$E = \{1, 2, 3, 4, 5\}$$

$$P(E) = \frac{5}{6}$$

Example-3:

Three coins are tossed once. Find the probability of getting:

- a. Three heads
- b. Two heads
- c. At least two heads
- d. At most two heads
- e. No head
- f. Three tails
- g. Exactly two tails
- h. No tail
- i. At most two tails

Example-3

Let $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$,
 $n(S)=8$

a. $E = \text{get three heads} = \{HHH\}$, $n(E)=1$

$$P(E) = \frac{1}{8}$$

b. $E = \text{get two heads} = \{HHT, HTH, THH\}$, $n(E)=3$

$$P(E) = \frac{3}{8}$$

c. $E = \text{get at least two heads} = \{HHT, HHT, HTH, THH\}$,
 $n(E)=4$

$$P(E) = \frac{4}{8} = \frac{1}{2}$$

Example-3:

d. $E = \text{At most two heads} = \{\text{HHT}, \text{HTH}, \text{HTT}, \text{THH}, \text{THT}, \text{TTH}, \text{TTT}\}$, $n(E)=7$

$$P(E) = \frac{7}{8}$$

e. $E = \text{get no head} = \{\text{TTT}\}$, $n(E)=1$

$$P(E) = \frac{1}{8}$$

f. $E = \text{get three tails} = \{\text{TTT}\}$, $n(E)=1$

$$P(E) = \frac{1}{8}$$

Example-3:

g. $E = \text{Exactly two tails} = \{HTT, THT, TTH\}$, $n(E)=3$

$$P(E) = \frac{3}{8}$$

h. $E = \text{get no tail} = \{HHH\}$, $n(E)=1$

$$P(E) = \frac{1}{8}$$

i. $E = \text{get at most two tails} = \{HHH, HHT, HTH, HTT, THH, THT, TTH\}$, $n(E)=7$

$$P(E) = \frac{7}{8}$$

Example-4:

If $\frac{2}{11}$ is the probability of an event A, what is the probability of event Not A?

$$\text{Let } P(A) = \frac{2}{11}$$

$$\begin{aligned} n(S) &= 11 \text{ and } n(A) = 2 \\ A' &= S - A, \\ \text{hence } n(A') &= n(S) - n(A) \\ &= 11 - 2 = 9 \end{aligned}$$

$$P(A') = \frac{9}{11}$$

$$\begin{aligned} P(A') &= 1 - P(A) \\ P(A') &= 1 - \frac{2}{11} \end{aligned}$$

$$P(A') = \frac{11-2}{11} = \frac{9}{11}$$

Example-5:

A letter is chosen at random from the word 'Assassination'. Find the probability that (i) the letter is vowel, (ii) the letter is consonant?

Given word: ASSASSINATION

$$n(S) = 13$$

Let A = set of vowel = {A, A, I, A, I, O}, $n(A) = 6$

B = set of consonant = {S, S, S, S, N, T, N}, $n(B) = 7$
 $= n(S) - n(A) = 13 - 6 = 7$

$$P(A) = \frac{6}{13}, \quad P(B) = \frac{7}{13}$$